Differentiating Powers of x

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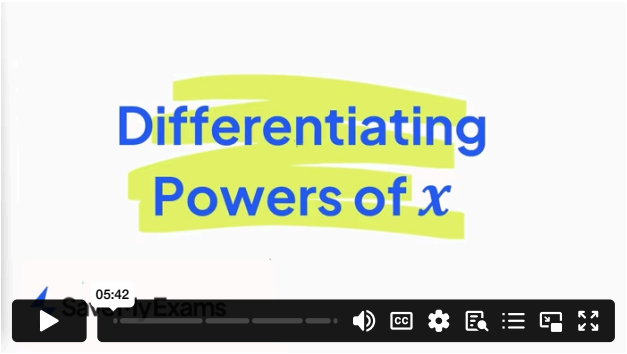

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Help me understand

Differentiating powers of x

How do I differentiate expressions with powers of x?

- Powers of x are differentiated according to the following formula
  - $\frac{d}{dx}(x^n) = nx^{n-1}$
    - Where n is any real constant
  - This can be written as:
    - If $f(x) = x^n$ then $f'(x) = nx^{n-1}$
    - If $y = x^n$ then $\frac{dy}{dx} = nx^{n-1}$

Examiner Tips and Tricks

$f'(x)$ is usually used when a function, $f(x)$, has been defined. $\frac{dy}{dx}$ is used when a graph of a function is given in the form $y = ...$ They both represent the derivative of an expression.

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- $\frac{d}{dx}(ax^n) = anx^{n-1}$
  - Where n and a are any real constants

- Differentiating terms with positive integer powers of x is straightforward

$$f(x) = x^2 \longrightarrow f'(x) = 2x^{2-1} = 2x$$

$$f(x) = 5x^3 \longrightarrow f'(x) = 5 \times 3x^{3-1} = 15x^2$$

$$y = -4x^5 \longrightarrow \frac{dy}{dx} = -4 \times 5x^{5-1} = -20x^4$$

$$y = \frac{2}{3}x^7 \longrightarrow \frac{dy}{dx} = \frac{2}{3} \times 7x^{7-1} = \frac{14}{3}x^6$$

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- But negative and fractional powers of x can also be differentiated this way

$$f(x) = 2x^{-1} \longrightarrow f'(x) = 2 \times (-1)x^{-1-1} = -2x^{-2}$$

$$f(x) = 5\sqrt{x} = 5x^{\frac{1}{2}} \longrightarrow f'(x) = 5 \times \frac{1}{2}x^{\frac{1}{2}-1} = \frac{5}{2}x^{-\frac{1}{2}} = \frac{5}{2\sqrt{x}}$$

$$\sqrt{x} = x^{\frac{1}{2}}$$

$$y = \frac{4}{5}x^{\frac{2}{3}} \longrightarrow \frac{dy}{dx} = \frac{4}{5} \times \frac{2}{3}x^{\frac{2}{3}-1} = \frac{8}{15}x^{-\frac{1}{3}}$$

$$y = \frac{1}{x^4} = x^{-4} \longrightarrow \frac{dy}{dx} = -4x^{-4-1} = -4x^{-5} = -\frac{4}{x^5}$$

$$\frac{1}{x^p} = x^{-p}$$

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- Don't forget these two special cases:
  - The derivative of a **linear function** is a **constant**
    - $\frac{d}{dx}(ax) = a$
    - Where a is any real constant
  - The derivative of a **constant function** is **zero**
    - $\frac{d}{dx}(a) = 0$
    - Where a is any real constant

- These allow you to differentiate constants and linear terms in x

$$f(x) = 3 \longrightarrow f'(x) = 0$$

$$f(x) = -5x \longrightarrow f'(x) = -5$$

$$y = \sqrt{7} \longrightarrow \frac{dy}{dx} = 0$$

$$y = x \longrightarrow \frac{dy}{dx} = 1$$

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- If the expression involves a sum or difference of terms just differentiate one term at a time

IS ZERO

$$y = x^4 - \sqrt{x} + \frac{4}{x}$$

$$\downarrow \qquad \downarrow \qquad \downarrow$$

$$\frac{dy}{dx} = 4x^3 - \frac{1}{2\sqrt{x}} - \frac{4}{x^2}$$

IN GENERAL IF $y = f(x) + g(x) + h(x) + \ldots$
THEN $\frac{dy}{dx} = f'(x) + g'(x) + h'(x) + \ldots$

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Examiner Tips and Tricks

- Take extra care when differentiating negative and fractional powers of **x** as mishandling negative signs and fractions is a common way to lose marks in these sorts of questions.

Worked Example

Teacher only

Fresh, expert-written questions
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